

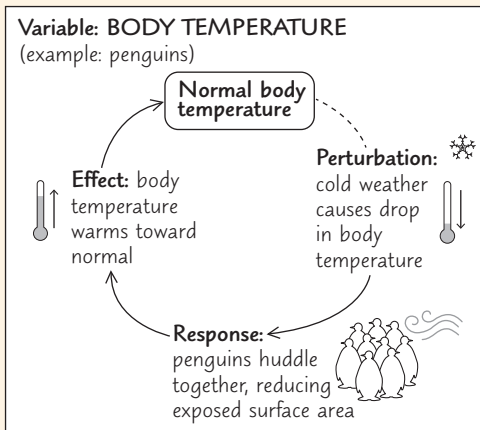
40 Basic Principles of Animal Form and Function

KEY CONCEPTS

- 40.1** Animal form and function are correlated at all levels of organization *p. 874*
- 40.2** Feedback control maintains the internal environment in many animals *p. 881*
- 40.3** Homeostatic processes for thermoregulation involve form, function, and behavior *p. 884*
- 40.4** Energy requirements are related to animal size, activity, and environment *p. 889*

Study Tip

Draw a diagram: When you encounter an example in the chapter of how an animal maintains a steady internal state, draw a simple circuit diagram (see example—illustrations are optional!). Label the variable being controlled, a perturbation that affects the variable, the response, and its effect in restoring the normal state.



Go to Mastering Biology

For Students (in eText and Study Area)

- Get Ready for Chapter 40
- Figure 40.17 Walkthrough: Thermoregulation in Humans

For Instructors to Assign (in Item Library)

- Everyday Biology: How to Keep Your Cool
- Tutorial: Thermoregulation

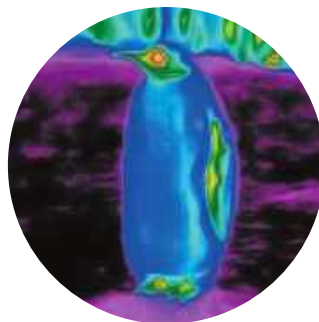


Figure 40.1 Emperor penguins (*Aptenodytes forsteri*) live in Antarctica, Earth's coldest and windiest continent. In summer, these birds catch fish by diving down 500 meters in water only 2°C above freezing. In winter, the females forage and the males incubate eggs while temperatures drop to -40°C and winds gust to 200 km/hr.

How do animals regulate their internal state even in changing or harsh environments?

Adaptations in **form, function, and behavior** help maintain an animal's internal environment. Adaptations that limit variation in temperature and other internal variables are widespread and diverse. Consider, for example, three adaptations that help an Emperor penguin stay warm:

Form (anatomy): An insulating layer of fat (blubber) reduces heat loss from most of the penguin's body (blue body areas in this thermal image).



Function (physiology): Rapid cycles of muscle contraction and relaxation during shivering produce heat at a cellular level.



Behavior: By packing together in groups of up to several thousand, Emperor penguins greatly reduce their exposure to wind and cold.

